

Epistemic Uncertainty in LLM Hallucinations

A Multi-Baseline Hallucination Detection Pipeline

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The Hallucination Problem Formulation

- **The Challenge:** Large language models (LLMs) often generate fluent but factually incorrect content, known as hallucinations.
- **Real World Impact:** Detecting these automatically is absolutely critical for safe deployment in high-stakes domains like medicine and law.
- **Project Hypothesis:** Hallucinated answers are associated with higher epistemic uncertainty, which can be approximated through token-level likelihood behavior under a causal language model.
- **The Mechanism:** When a model generates tokens outside its reliable knowledge boundary, its internal token-level probabilities become measurably less confident.



Dataset & Problem Formulation



The Benchmark

We utilized the **HaluEval QA** benchmark dataset, consisting of paired factual and hallucinated candidate answers.



Classification Task

Framed as a binary classification problem. The system evaluates a question and answer pair: Factual (Label 0) or Hallucinated (Label 1).



Data Scaling

Expanded to 15,998 training samples and 2,002 validation samples, maintaining perfect class balance for robust evaluation.

Initial Implementation

Dual-Pathway Processing: Simultaneously extracts text patterns and model confidence from the input pair.

Pathway 1 (Text Features): Uses TF-IDF to create a sparse matrix of up to 20,000 word-level features.

Pathway 2 (Uncertainty Features): Uses distilgpt2 to extract 10 numeric confidence metrics.

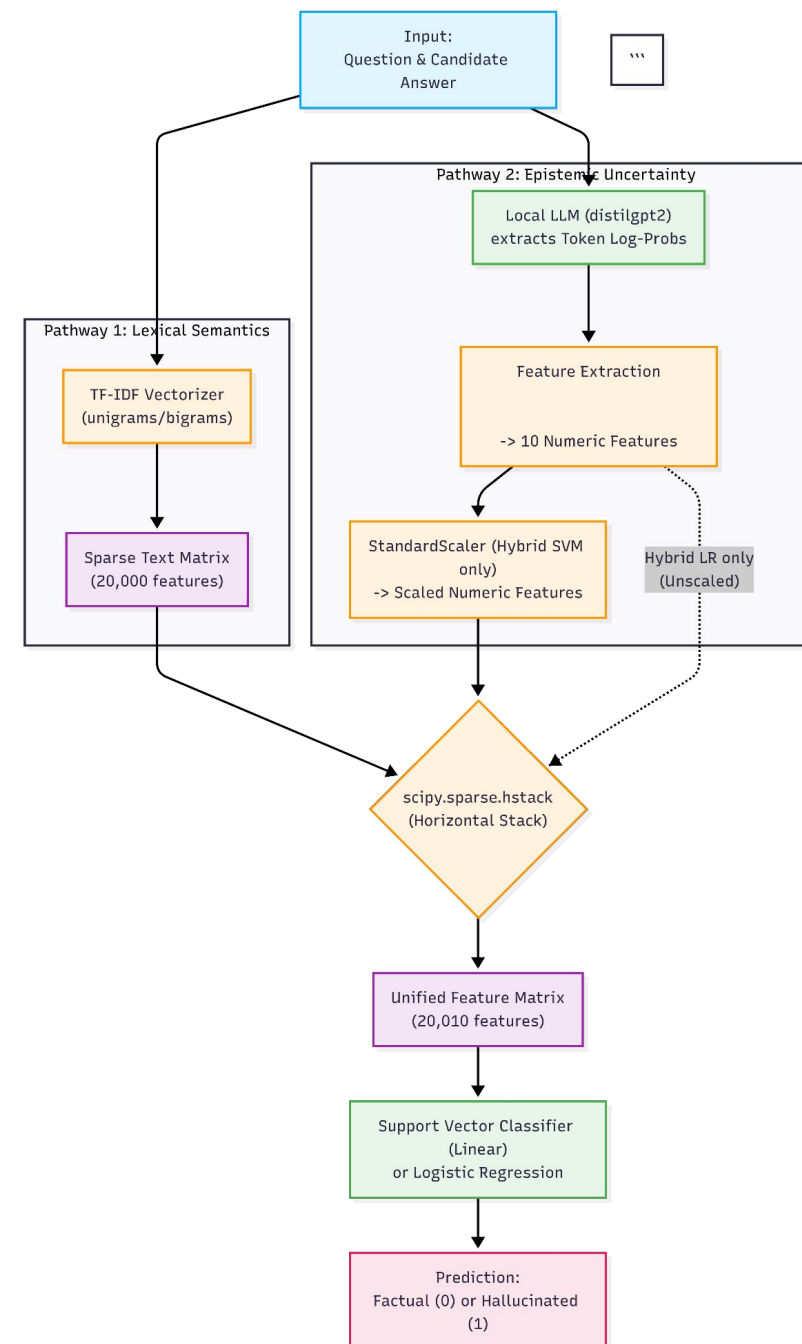
Feature Scaling Difference:

Hybrid SVM: Normalizes the 9 numeric metrics so they blend smoothly with the text data.

Hybrid LR: Uses unscaled, raw metrics

Feature Merging: Both pathways are horizontally stacked into a single, unified matrix of 20,010 features.

Classification Output: The final matrix is fed into a classifier to predict whether the answer is Factual (0) or Hallucinated (1).



Detection Paradigms



Lexical

TF-IDF Lexical SVM.



Semantic

Semantic SVM



Uncertainty

Entropy Base
+
Entropy Upgraded



Evidence / Retrieval

RAG Compare
+
NLI Evidence

Hybrid Models

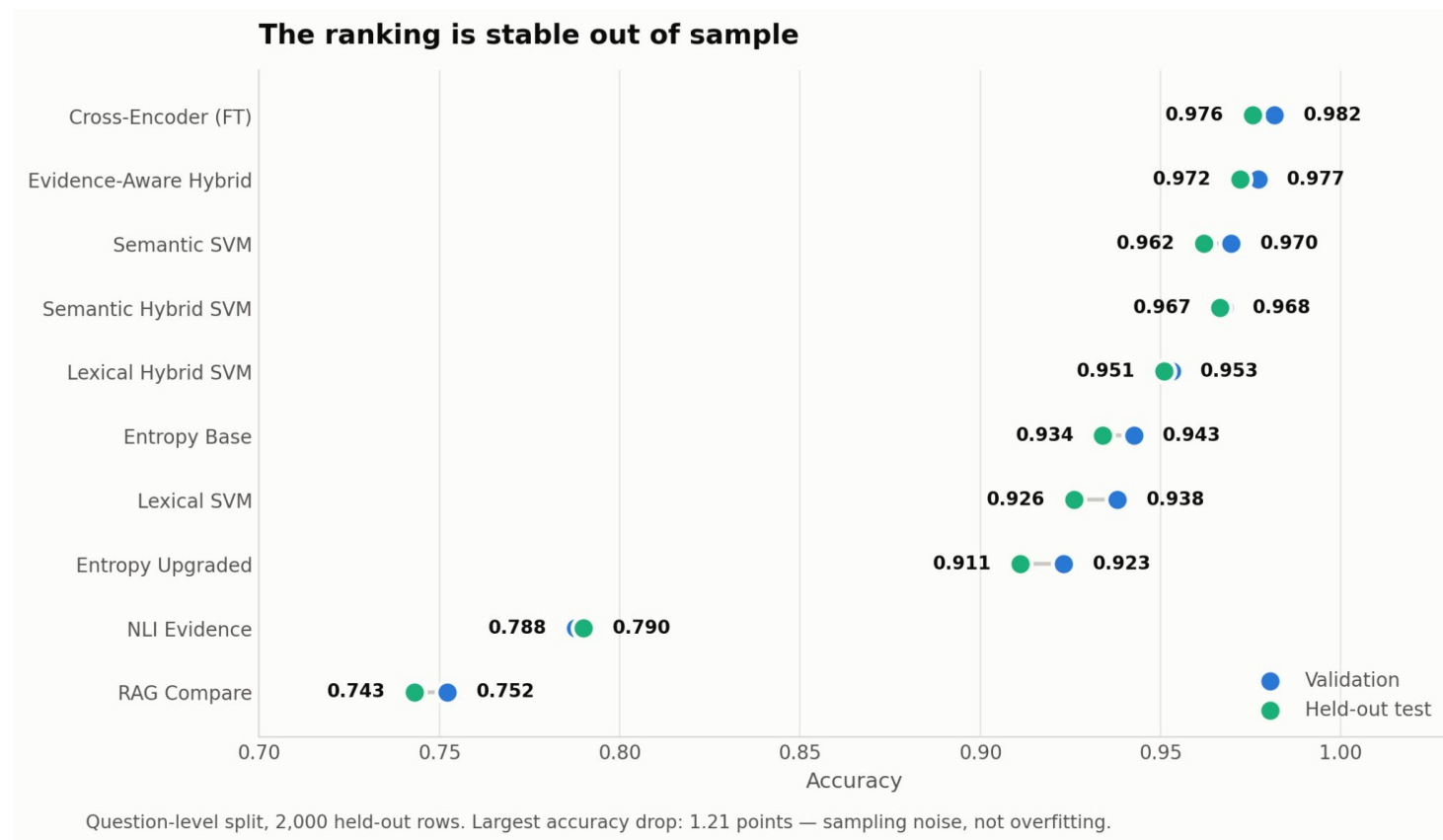
Model Architecture	Features	Purpose
Lexical Hybrid SVM	TF-IDF + Uncertainty (log-prob)	Are Lexical & Uncertainty signals <u>complementary</u> ?
Semantic Hybrid SVM	Semantic interaction + Uncertainty	Adds <u>uncertainty</u> on top of the strongest baseline
Evidence-Aware Hybrid	Semantic + Uncertainty + Context-shift + NLI	Combines <u>all four</u> paradigms
Cross-Encoder (FT)	Fine-tuned cross-encoder reading q+a jointly	Supervised <u>upper-bound</u> baseline

Scaled Experimental Results

Model Architecture	Accuracy	Macro F1	AUROC
Lexical SVM	0.9381	0.9380	0.9729
Entropy Base	0.9426	0.9425	0.9765
Entropy Upgraded	0.9231	0.9231	0.9654
Semantic SVM	0.9695	0.9695	0.9870
NLI Evidence	0.7877	0.7866	0.7943
RAG Compare	0.7522	0.7490	0.7723
Lexical Hybrid SVM	0.9530	0.9530	0.9805
Semantic Hybrid SVM	0.9675	0.9675	0.9878
Evidence-Aware Hybrid	0.9770	0.9770	0.9951
Cross-Encoder (FT)	0.9815	0.9815	0.9895

Held-out Test Evaluation

- Ran the full ten-baseline matrix on the disjoint 2,000-row test split using the same trained models and thresholds.
- Every method's accuracy and AUROC lands within 1.3 points of its validation value (largest drop: Lexical SVM and Entropy Upgraded, -1.21).
- NLI Evidence and the Evidence-Aware Hybrid actually score marginally higher on test.



Ablation Study & Feature Contributions

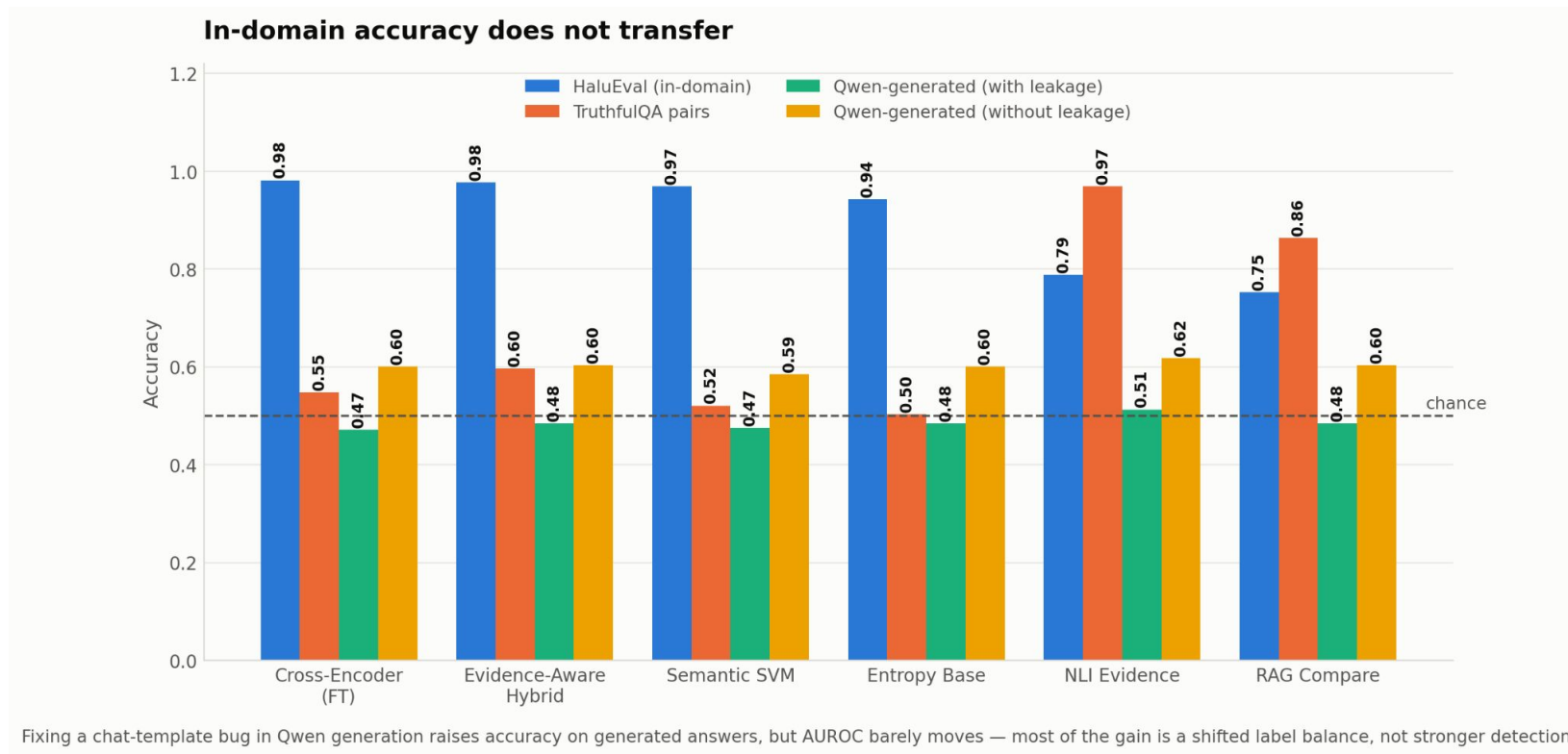
Setting	Features	Accuracy	Macro F1	AUROC
lexical_only	TF-IDF	0.9391	0.9390	0.9703
uncertainty_only	log-probability features	0.7338	0.7152	0.9740
hybrid_no_context	TF-IDF + log-probs	0.9446	0.9446	0.9800
hybrid_with_context	TF-IDF + log-probs + Δ	0.9735	0.9735	0.9919

Uncertainty is miscalibrated, not uninformative - alone it reaches 0.9740 AUROC but only 0.7152 macro F1, pushing 1,512 of 2,002 examples to "hallucinated" (522 false positives) while recall stays at 990/1,001. The bottleneck is the decision threshold, not the ranking signal.

Signals are complementary - lexical + uncertainty $\rightarrow$ 0.9800 AUROC, above either alone

Grounding is the largest single gain - adding context-delta features increases 0.9800 $\rightarrow$ 0.9919 AUROC

Transfer Evaluation



Inverted ranking: the weakest in-domain methods (NLI Evidence, RAG Compare) are the only ones that survive. Evidence grounding generalises; memorised benchmark artifacts do not.

Total collapse on generated answers: Entropy Base, Evidence-Aware Hybrid and RAG Compare all report exactly 0.4847 because each predicts every one of the 817 examples as hallucinated — 396/817 is precisely what any constant classifier scores here.

Caveat: the Qwen-generated set was labeled automatically with NLI; our audit found prompt leakage and many low-margin decisions. Read it as indicative, not clean ground truth.

Error Analysis & Limitations



Overconfident Signals

RAG Compare / NLI keep errors even at high decision margins



Short Answer is Hard

Error rates peak in the shortest answer bucket across nearly all baselines



Hybrid's Fail

The Evidence-Aware Hybrid has a much flatter high-confidence error profile. Signals work best as complementary constraints, not standalone rules.

Conclusion

Uncertainty is complementary, not sufficient

Token-level log-probabilities from distilgpt2 carry real signal — 0.974 AUROC alone — but only 0.715 macro F1. They rank well and decide badly. They earn their place fused with semantic and evidence features, not on their own.

Change beats level

How a model's confidence shifts once evidence is supplied is more informative than its raw confidence. Adding context-delta features was the single largest ablation gain: 0.9800 → 0.9919 AUROC.

Joint models win, but not the ones we expected

Best accuracy comes from the fine-tuned Cross-Encoder (0.9815); best ranking from the Evidence-Aware Hybrid (0.9951). Semantic and evidence signals do more work than lexical overlap.

The benchmark is not the task

0.98 in-domain collapses to near-chance on open-domain generated answers. The methods that transfer are the evidence-grounded ones that scored worst on HaluEval. The contribution is diagnostic: where cheap uncertainty helps, where it fails, and why.

Thank You!

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Dataset

HaluEval QA (Li et al., EMNLP 2023)

TruthfulQA (Lin et al., ACL 2022)

Methodology

10 baselines, 4 hybrids, ablations, held-out test, 2 transfer evaluations

Scoring Model

distilgpt2 (82M)

Qwen2.5-1.5B-Instruct

Github Repository

<https://github.com/TAtaTrkylmz/CENG463-FinalProject>

Image Sources



<https://siliconangle.com/wp-content/blogs.dir/1/files/2023/12/ChatGPT-hallucination.jpg>

Source: siliconangle.com
